



ST BENEDICT'S

SUBJECT	Physical Science Paper 2	DATE	23 November 2020
GRADE	11	MARKS	200
EXAMINER	Mr du Toit	MODERATOR	Mr McCoy
NAME	MARKING GUIDELINES	DURATION	3 hours
CLASS			

QUESTION 1 – MULTIPLE CHOICE

20 MARKS

1.1	A ✓✓	(2)
1.2	C ✓✓	(2)
1.3	B ✓✓	(2)
1.4	C ✓✓	(2)
1.5	C ✓✓	(2)
1.6	B ✓✓	(2)
1.7	D ✓✓	(2)
1.8	C ✓✓	(2)
1.9	C ✓✓	(2)
1.10	A ✓✓	(2)

QUESTION 2 NTOMBELA**21 MARKS**

2.1

2.1.1	E✓, F✓
2.1.2	A✓, B✓, D✓
2.1.3	E✓, F✓, G✓
2.1.4	D✓
2.1.5	C✓

(10)

Minus 1 for every error

2.2.1 a bond which occurs between atoms✓ within molecules✓ (2)

2.2.2 Polar✓ covalent ✓ (bond) (2)

2.2.3 There is an electronegativity difference between chlorine and hydrogen (END>0) OR chlorine has a higher electronegativity than hydrogen ✓
The chlorine atom attracts the **bonding pair of electrons**✓
more than the hydrogen atom. (comparison) ✓ (3)

2.2.4 weak force of attraction ✓ between molecules, ions, or atoms of noble gases✓ (2)

2.2.5 Dipole-dipole force ✓ (1)

2.2.6 δ^+ ✓ (1)

QUESTION 3 JOOSTE

20 MARKS

- 3.1 the loss of electrons ✓✓ AON (2)
- 3.2 $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$ ✓ (2)
- 3.3 a substance that accepts electrons. ✓✓ AON (2)
- 3.4 Fluorine ✓ (1)
- 3.5.1 Aqueous✓ or molten (liquid)✓ state
Ions free to move ✓ (3)
- 3.5.2 Ionic bond OR electrostatic force/ attraction✓ (1)
- 3.5.3 Many ✓
strong ionic **bonds** ✓ OR strong electrostatic **forces**
More energy required to **break** bonds OR overcome electrostatic **forces**✓ (3)
- 3.6.1 As the atomic radius (of the halide/ compound size) increases ✓ the melting point (of the sodium halide/ compound) decreases ✓ (order matters -1 if order switched) (2)
- 3.6.2 A greater radius means greater distance ✓
between ions/ charges ✓
Weaker the electrostatic forces ✓ (or ionic BONDS)
Less energy required to overcome weaker electrostatic forces ✓ (or BREAK bonds) (4)

QUESTION 4 NTOMBELA (4.1 – 4.3) AND JOOSTE (4.4 – 4.7)**30 MARKS**

4.1 the net change ✓ of chemical potential energy of the system ✓

(2)

4.2.1 and 4.2.2

I	Energy of reactants ✓
II	Activation energy ✓
III	Energy released ✓
IV	Energy of products ✓
A	Reactants ✓
B	Reaction coordinate ✓

(6)

4.3.1 Activated complex ✓

(1)

4.3.2 Collide✓

with sufficient **kinetic** energy ✓

and correct orientation ✓

(3)

4.3.3 Bonds are breaking ✓ and forming✓

(2)

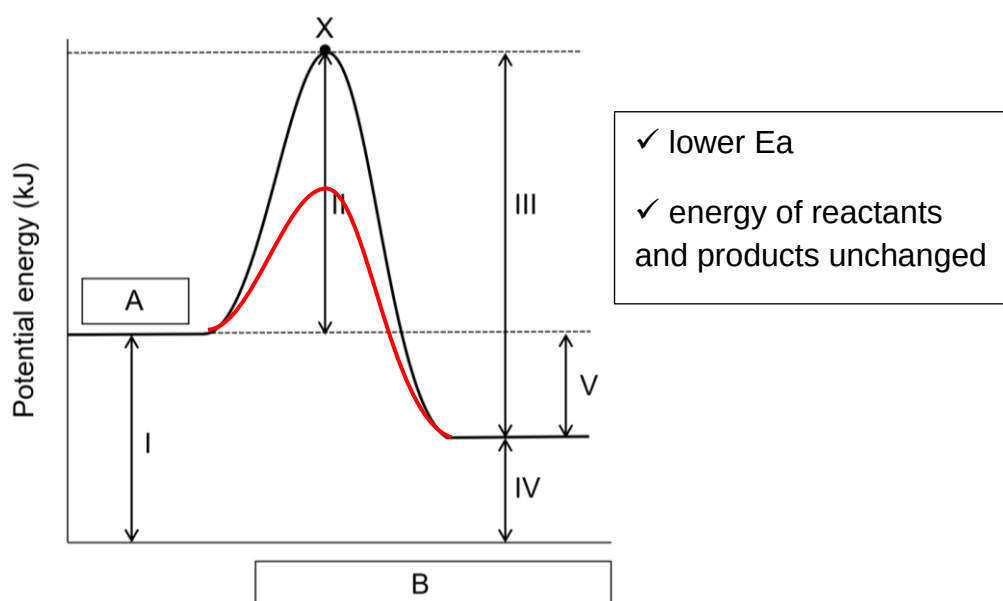
4.4 Exothermic. ✓

The energy released is greater than energy absorbed ✓ OR energy of products is less than reactants) ✓

There is a net loss/ release in energy (or change in enthalpy < 0) ✓

(3)

4.5



(2)

4.6 No effect ✓

The energy of products and reactants ✓ not affected ✓ (catalyst lowers E_a one mark only) (3)

4.7.1

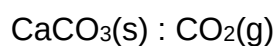
$$n = \frac{V}{Vm} \checkmark \text{ (can be implied)}$$

$$n = \frac{0,5}{22,4} \checkmark$$

$$n = 0,02 \text{ mol} \checkmark$$

(3)

4.7.2



1:1 ✓ (using ratio)

$$n = 0,02 \text{ mol of CaCO}_3$$

$$n = \frac{m}{M} \checkmark \text{ (can be implied)}$$

$$0,02 = \frac{m}{100} \checkmark$$

$$m = 2,23 \text{ g}$$

$$\% \text{purity} = \frac{\text{mass}(\text{pure})}{\text{mass}(\text{impure})} \times 100 \checkmark \text{ (can be implied)}$$

$$= \frac{2,23}{3} \checkmark = 74,40\% \checkmark$$

(5)

QUESTION 5 MCCOY

29 MARKS

5.1 the change in concentration per unit time✓ of either a reactant or product. ✓ (2)

5.2 $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$ ✓ (3)

5.3.1 surface area of zinc✓ (1)

5.3.2 reaction rate✓ (1)

5.3.3 temperature, concentraton of sulphuric acid , amount of zinc, equal time period (ANY two) ✓✓ (2)

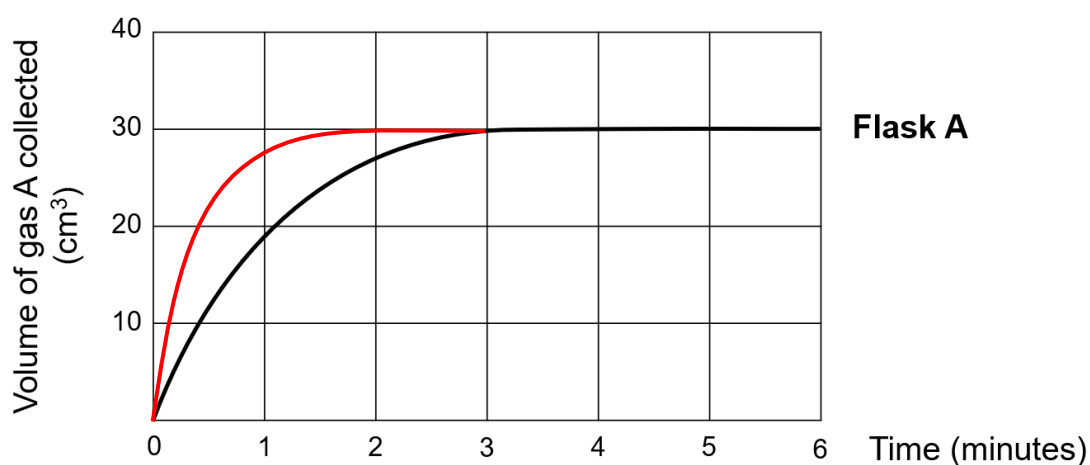
5.4 Both affect reaction rate. ✓
Keeps the test fair ✓ (2)

5.5 No ✓
Hydrogen is a light gas ✓
Decrease in mass would be negligible WTTF ✓ (3)

5.6 Average rate = $30/3$ ✓ = 10 ✓ $\text{cm}^3 \cdot \text{min}^{-1}$ ✓ OR $30/180 = 1,6 \times 10^{-4} \text{ dm}^3 \cdot \text{s}^{-1}$ (3)

5.7 Gradient represents reaction rate ✓
The concentration of acid decreases✓ (reactants getting used up) (2)

5.8



Starts at zero ✓

Steeper gradient and ends sooner ✓

Ends at 30 cm^3 ✓ (3)

5.9 Powdered zinc solid has a greater surface area ✓
More particles exposed for collisions ✓
More collisions ✓
More effective collisions **per unit time** ✓ (per unit time at either place) (4)

5.10 Correcting statement:

the amount of product produced will NOT always be greater as there will be moles of solute to react to form product. ✓

There will be more moles of solute as a whole ✓

But not per unit volume✓ or only when volume of limiting reagent increased ✓ (3)

QUESTION 6 MCCOY**28 MARKS**

6.1.1

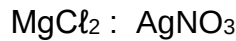
(6)



$$n = c \cdot V \checkmark = (0,25)(0,055 \checkmark) = 0,01375 \text{ mol} \checkmark$$



$$n = c \cdot V = (0,15)(0,070) = 0,0105 \text{ mol} \checkmark$$

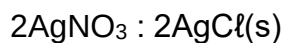


1 : 2

$$n = \frac{0,0105}{2} = 5,25 \times 10^{-3} \text{ of MgCl}_2 \text{ required} \checkmark$$

If AgNO_3 is limiting then $5,25 \times 10^{-3} \text{ mol}$ of MgCl_2 required. 0,01375 available (enough).
 AgNO_3 limiting. $\checkmark$

6.1.2

2 : 2 $\checkmark$

$$n = 0,0105 \text{ mol}$$

$$n = \frac{m}{M}$$

$$0,0105 = \frac{m}{143,5}$$

$$m = 1,51 \text{ g answer} \checkmark \text{ correct rounding} \checkmark$$

Unrounded: 1,44 g

(3)

6.1.3

$$\% \text{yield} = \frac{\text{actual}}{\text{theoretical}} = \frac{1,49 \checkmark}{1,51 \checkmark} \times 100 = 98,98\% \checkmark$$

(3)

6.2.1 They have known concentrations $\checkmark$

(1)

6.2.2 the substance in which another substance is dissolved $\checkmark$, forming a solution $\checkmark$

(2)

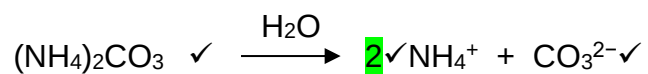
6.2.3 water or H_2O $\checkmark$

(1)

6.2.4 Mass out required solid $\checkmark$ Transfer all solute to volumetric flask $\checkmark$ Rinse with distilled water $\checkmark$ Make up to final volume with distilled water $\checkmark$

(4)

6.2.5



(3)

6.2.6

$$\text{NH}_4\text{OH}: n = c \cdot V = 0,2 \times 0,25 = 0,05 \text{ mol } \checkmark$$

$$(\text{NH}_4)_2\text{CO}_3: n = c \cdot V = 0,3 \times 0,5 = 0,15 \text{ mol } \times 2 = 0,3 \text{ mol } \checkmark$$

$$c = \frac{n}{V}$$

$$c = \frac{0,05 + 0,3\checkmark}{0,25 + 0,50\checkmark}$$

$$c = 0,47 \text{ mol} \cdot \text{dm}^{-3}\checkmark$$

(5)

QUESTION 7 DU TOIT

13 MARKS

- 7.1 an atom or a group of atoms ✓ that form the centre of chemical activity in the molecule ✓ (2)
- 7.2 R-O-H ✓ (1)
- 7.3 hydroxyl ✓ (1)
- 7.4 but ✓ ane ✓ -1,3- ✓ diol ✓ (4)
- 7.5.1 Hydrogen bond forces ✓ (1)
- 7.5.2 B ✓
more hydrogen bonds ✓ between molecules ✓ (B has two hydroxyl groups)
more energy required to overcome a larger number of hydrogen bond forces ✓ (4)

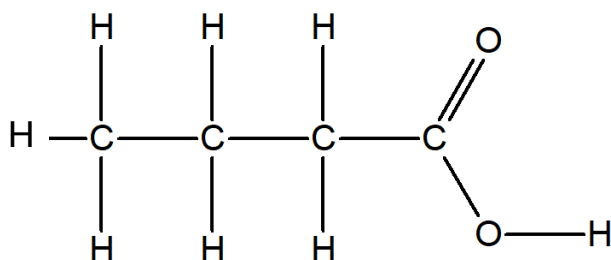
QUESTION 8 DU TOIT**17 MARKS**

8.1 ethyl ✓ butanoate ✓ (2)

8.2 ester ✓ (1)

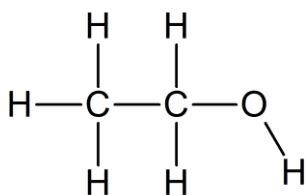
8.3 butan✓oic acid ✓ coe 8.1 (2)

8.4



✓Four carbons coe 8.1

✓Carboxyl group



✓Two carbons

✓Hydroxyl group

(4)

8.5 H_2SO_4 ✓✓ (only one mark for name) (2)

8.6 Compounds having the same molecular formula✓ but different structural formulae✓ (2)

8.7 butyl ethanoate ✓✓ (1 mark for structure) OR methyl pentanoate OR propyl propanoate OR pentyl methanolate coe 8.1 (2)

8.8 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$ ✓✓ (accept abbreviated version: $\text{CH}_3(\text{CH}_2)_4\text{COOH}$) coe 8.1 (2)

QUESTION 9 DU TOIT

22 MARKS

9.1 Contains only hydrogen and carbon ✓ **and** has at least one double bond between carbon atoms. ✓

(2)

9.2

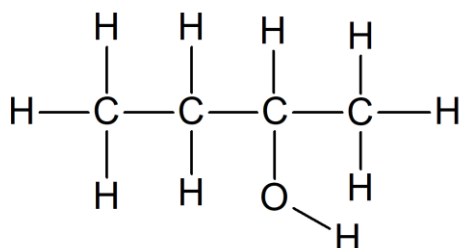
9.2.1 Elimination ✓

(1)

9.2.2 Elimination ✓

(1)

9.3



✓ Four carbons

✓ Hydroxyl on position 2

(2)

9.4 Concentrated alkali/ NaOH/ KOH (in ethanol solvent) ✓ and heat ✓

(2)

9.5.1 HCl ✓

(1)

9.5.2 H₂ ✓

(1)

9.6 C₄H₈ + 6O₂ → 4CO₂ + 4H₂O Reactants ✓ balanced ✓ products ✓

(3)

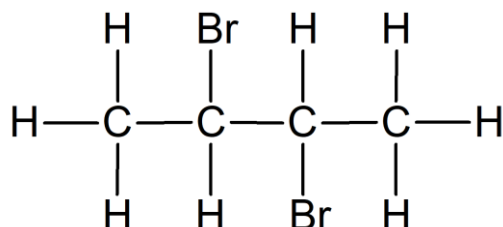
9.7.1 Rapid ✓ decolouring ✓

(2)

9.7.2 Halogenation ✓

(1)

9.7.3



✓ two bromides on position 2 and 3

✓ four carbons

(2)

9.7.4 2,3-dibromo✓butane✓ (coe from 9.7.3)

(2)

9.8.1 Thermal cracking ✓

(1)

9.8.2 C_nH_{2n} ✓

(1)